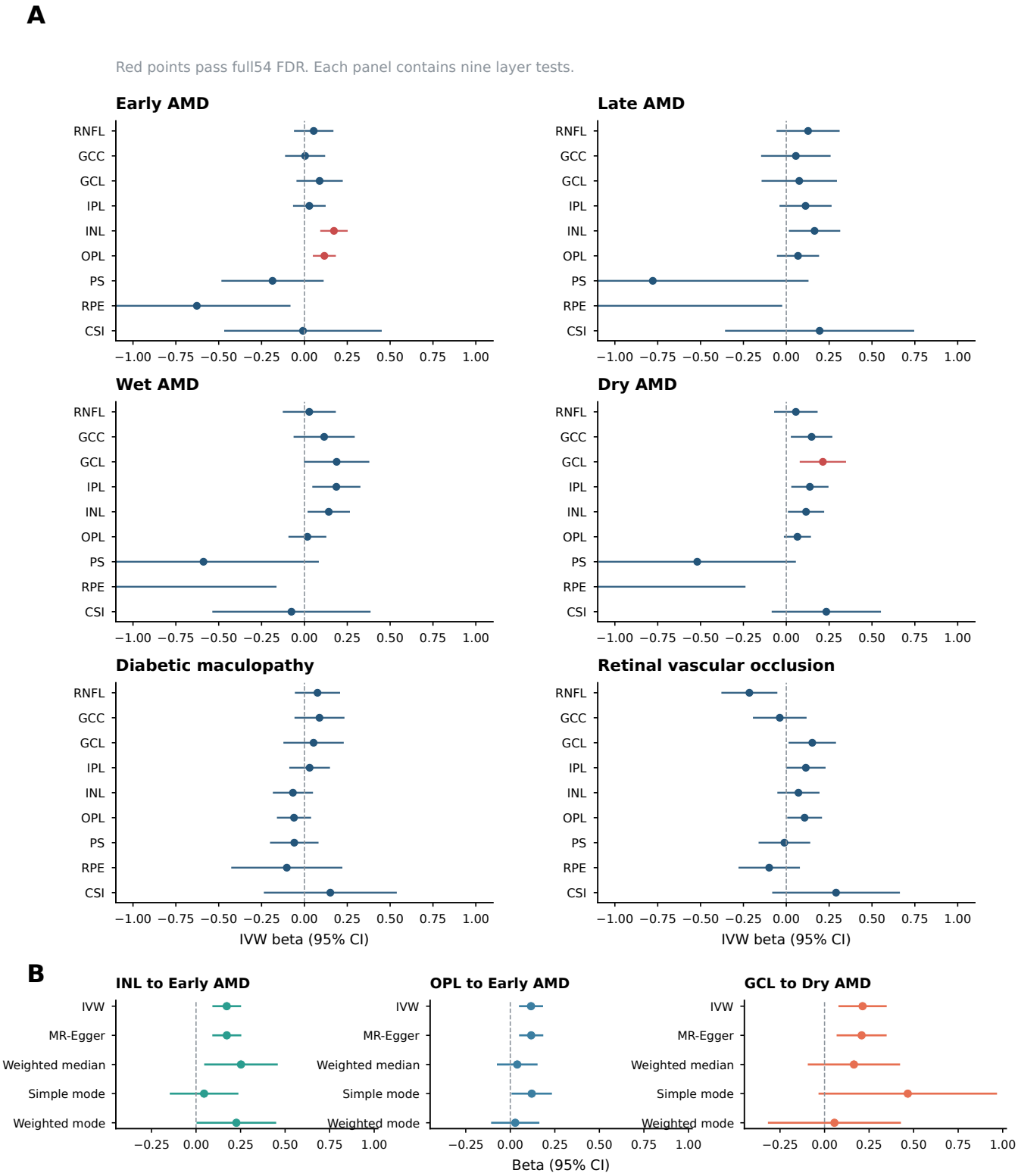
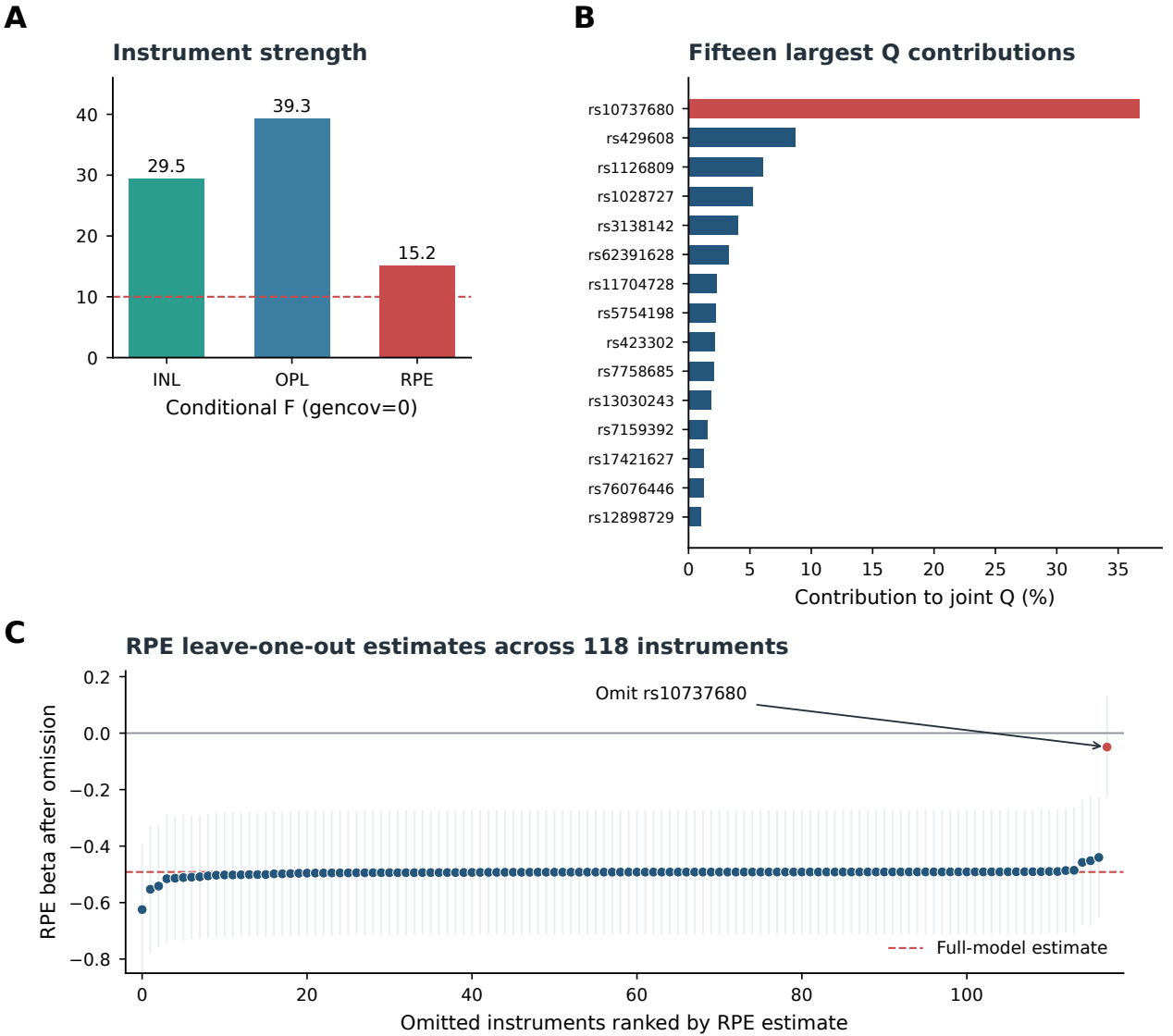
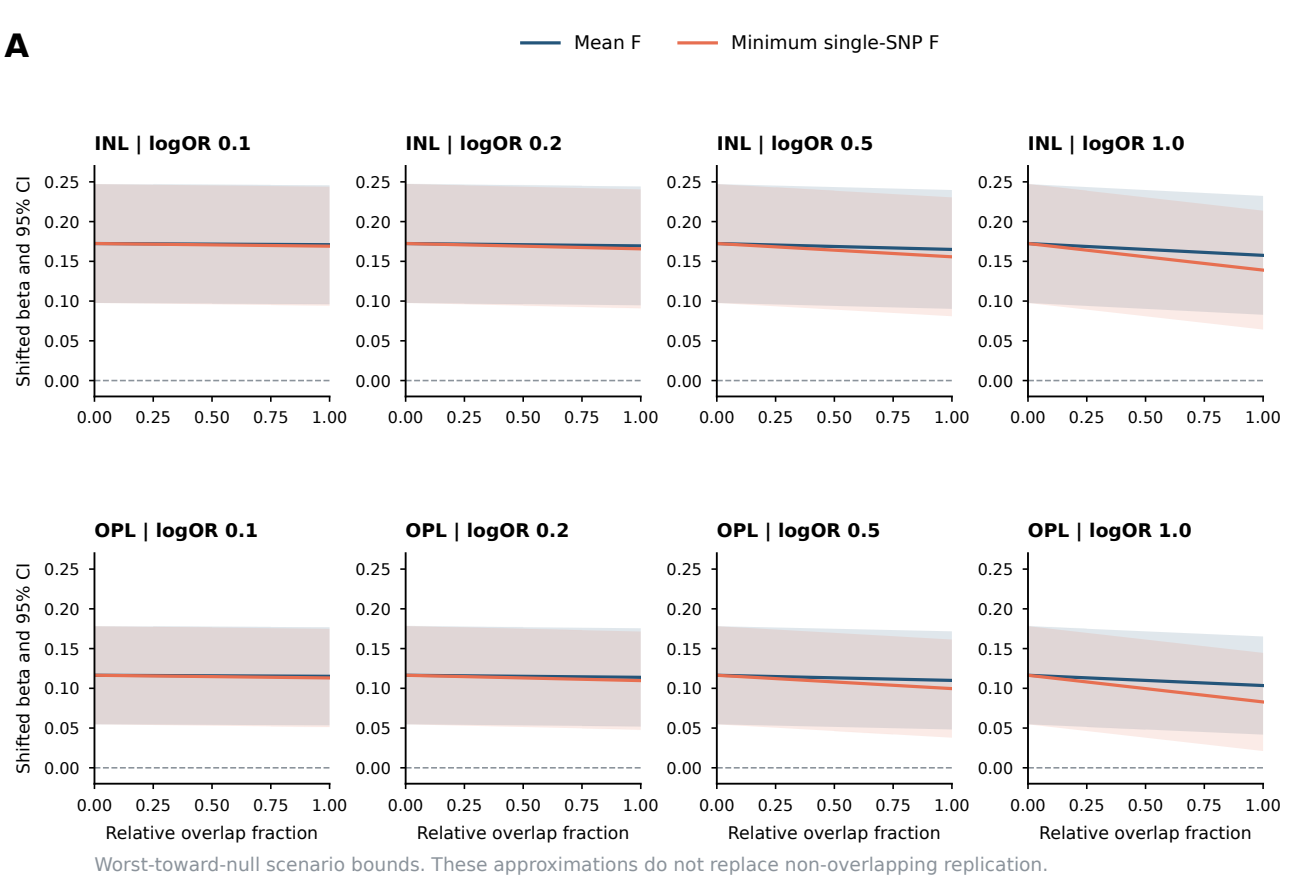
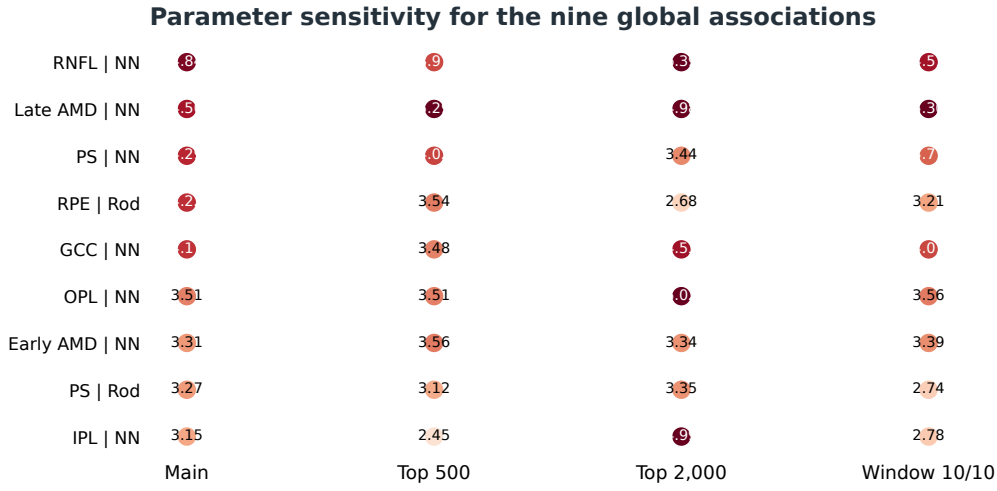

A
Red points pass full54 FDR. Each panel contains nine layer tests.
Early AMD
Late AMD
Wet AMD
Dry AMD
Diabetic maculopathy
Retinal vascular occlusion
RNFL
GCC
GCL
IPL
INL
OPL
PS
RPE
CSI
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
IVW beta (95% CI)
B
INL to Early AMD
OPL to Early AMD
GCL to Dry AMD
IVW
MR-Egger
Weighted median
Simple mode
Weighted mode
−0.25 0.00 0.25 0.50 0.75 1.00
Beta (95% CI)

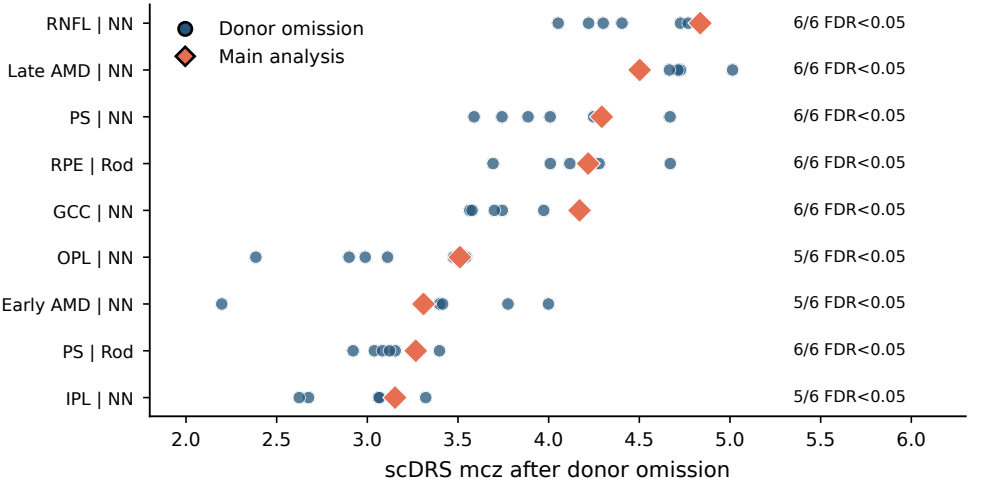

A
Instrument strength
40
30
20
10
0
29.5
39.3
15.2
INL
OPL
RPE
Conditional F (gencov=0)
B
Fifteen largest Q contributions
rs10737680
rs429608
rs1126809
rs1028727
rs3138142
rs62391628
rs11704728
rs5754198
rs423302
rs7758685
rs13030243
rs7159392
rs17421627
rs76076446
rs12898729
0
5
10
15
20
25
30
35
Contribution to joint Q (%)
C
RPE leave-one-out estimates across 118 instruments
Omit rs10737680
0.2
0.0
−0.2
−0.4
−0.6
−0.8
RPE beta after omission
Full-model estimate
0
20
40
60
80
100
Omitted instruments ranked by RPE estimate


A
Mean F
Minimum single-SNP F
INL | logOR 0.1
INL | logOR 0.2
INL | logOR 0.5
INL | logOR 1.0
OPL | logOR 0.1
OPL | logOR 0.2
OPL | logOR 0.5
OPL | logOR 1.0
Shifted beta and 95% CI
0.25
0.20
0.15
0.10
0.05
0.00
0.00
0.25
0.50
0.75
1.00
Relative overlap fraction
Worst-toward-null scenario bounds. These approximations do not replace non-overlapping replication.

A


Parameter sensitivity for the nine global associations
RNFL | NN
Late AMD | NN
PS | NN
RPE | Rod
GCC | NN
OPL | NN
Early AMD | NN
PS | Rod
IPL | NN
3.51
3.31
3.27
3.15
3.54
3.48
3.51
3.56
3.12
2.45
3.44
2.68
3.34
3.35
3.21
3.56
3.39
2.74
2.78
Main
Top 500
Top 2,000
Window 10/10

B


Leave-one-donor-out robustness within the same atlas
Donor omission
Main analysis
RNFL | NN
Late AMD | NN
PS | NN
RPE | Rod
GCC | NN
OPL | NN
Early AMD | NN
PS | Rod
IPL | NN
6/6 FDR<0.05
6/6 FDR<0.05
6/6 FDR<0.05
6/6 FDR<0.05
6/6 FDR<0.05
5/6 FDR<0.05
5/6 FDR<0.05
6/6 FDR<0.05
5/6 FDR<0.05
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
scDRS mcz after donor omission